

Government Health Spending per Person and Life Expectancy at Birth, 2020

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Analytical report based on country-level data for 2020

Executive Summary

This report examines the cross-country relationship between government health spending per person, measured in current US dollars, and life expectancy at birth in 2020. The analysis uses country-level observations from the two uploaded datasets and matches countries using their three-letter country codes.

The analytical sample includes 192 countries with non-missing values for both indicators and positive government health spending per person. The results show a clear positive association: countries that spend more public resources on health per person tend, on average, to have longer life expectancy.

The raw Pearson correlation between government health spending per person and life expectancy is 0.624. Because health spending per person is highly skewed, the report also assesses the relationship using the logarithm of spending; the correlation between log10 spending and life expectancy is 0.823. In a simple bivariate log-linear fit, a tenfold increase in government health spending per person is associated with approximately 6.6 additional years of life expectancy. This should be interpreted as an association, not as causal evidence.

1. Data and Methodological Approach

The input datasets were reviewed before analysis. Both datasets are organized in wide format: each row represents a country, while the columns contain annual values. The relevant column for this report is 2020.

1.1 Initial data review

Life expectancy dataset: 194 rows and 303 columns. Government health spending dataset: 192 rows and 25 columns.

Preview of life expectancy data:

geo	name	Life expectancy, 2020
afg	Afghanistan	64.60
ago	Angola	62.80
alb	Albania	77.40
and	Andorra	82.70
are	UAE	79.50
arg	Argentina	76.40
arm	Armenia	70.70
atg	Antigua and Barbuda	77.30

Preview of government health spending per person data:

geo	name	Government health spending per person, 2020 USD
afg	Afghanistan	6.12
ago	Angola	29.04
alb	Albania	157.86
and	Andorra	2,382.40
are	UAE	1,313.89
arg	Argentina	570.74
arm	Armenia	103.06
atg	Antigua and Barbuda	550.28

1.2 Data preparation checks

For mapping, the analysis created one 2020 record per country. The life expectancy map input contains 194 country rows, with 0 duplicate country codes and 0 missing 2020 values. The government health spending map input contains 192 country rows, with 0 duplicate country codes and 0 missing 2020 values. This confirms that both map inputs are in the correct one-row-per-country format.

2. Descriptive Mapping Results

The maps show the global distribution of the two indicators in 2020. They should be read as descriptive visualizations of country-level differences rather than as evidence of causal mechanisms.

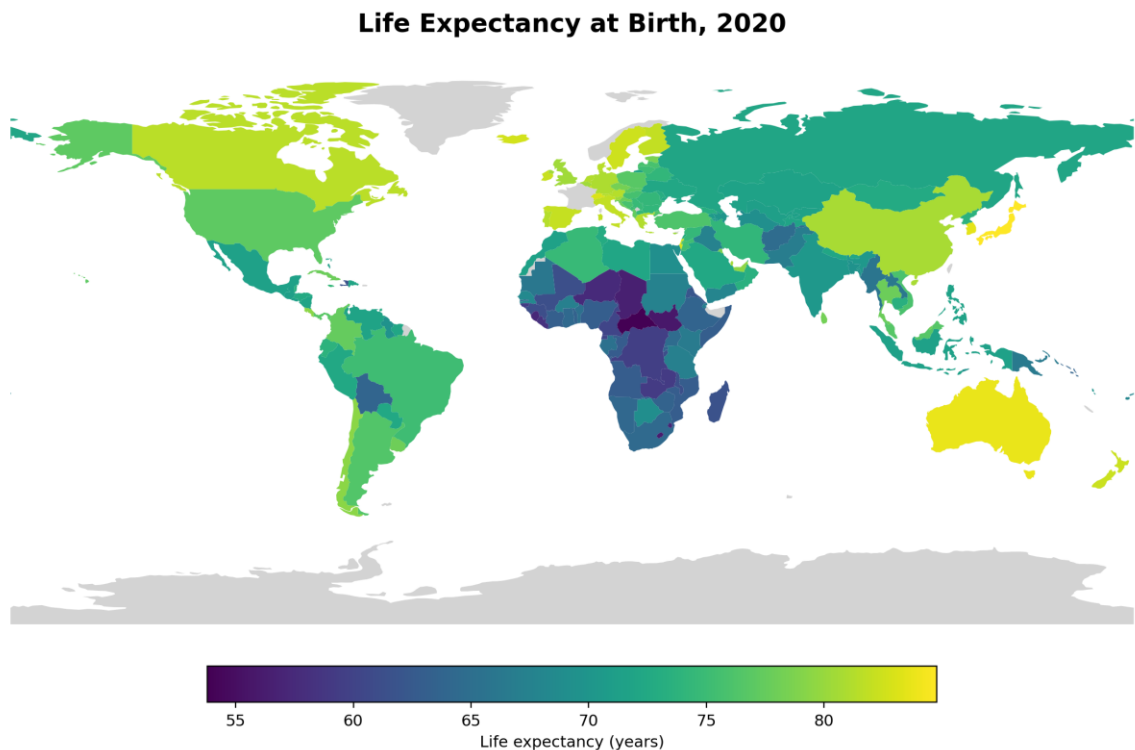


Figure 1. Life expectancy at birth, 2020

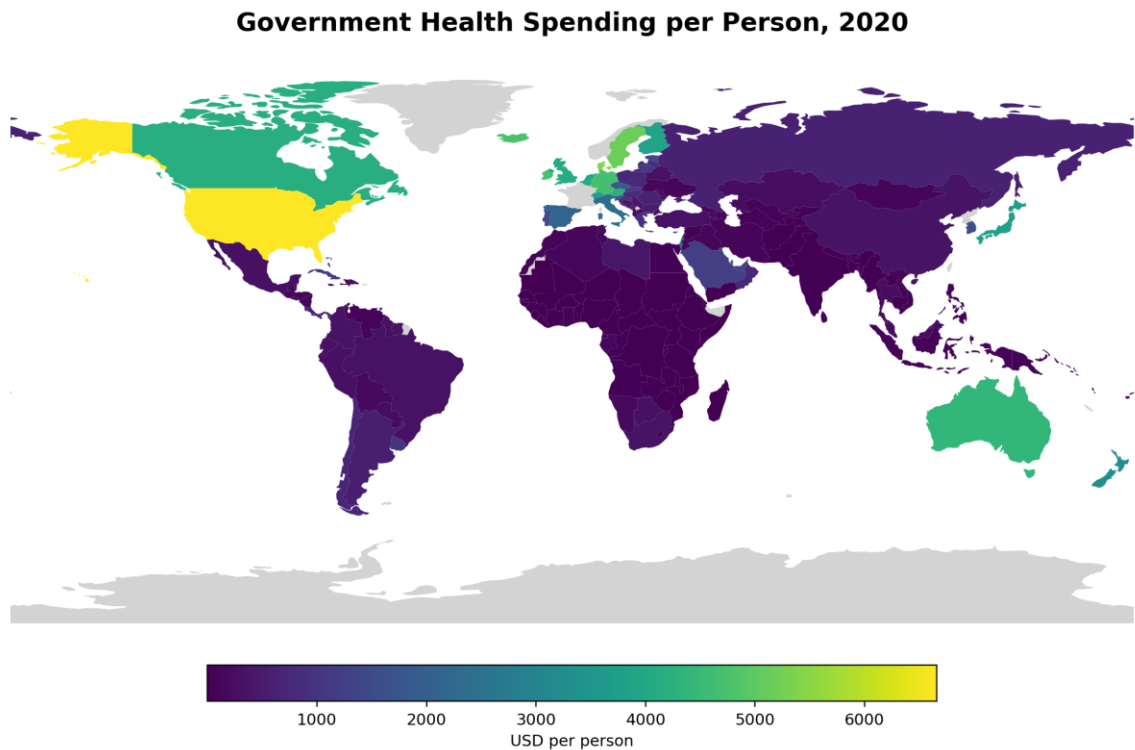


Figure 2. Government health spending per person, 2020

Government health spending and life expectancy, 2020

3. Relationship between Spending and Life Expectancy

The scatterplot compares countries according to their level of government health spending per person and life expectancy at birth in 2020. The horizontal axis is shown on a logarithmic scale to improve readability and because public spending per person varies by several orders of magnitude across countries.

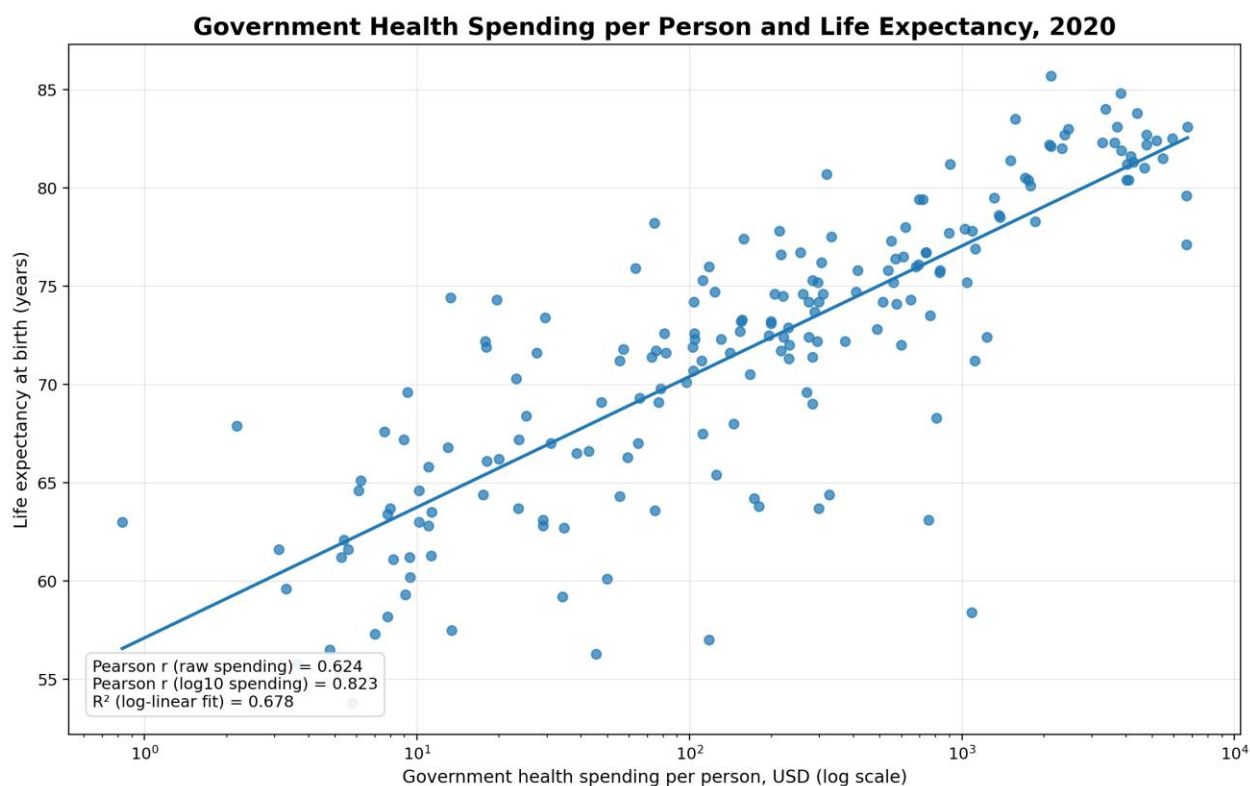


Figure 3. Relationship between government health spending per person and life expectancy, 2020

4. Health Economics Interpretation

The analysis is consistent with a central expectation in health economics: greater public investment in health is associated with better population health outcomes. Higher government health spending per person can support broader service coverage, better availability of essential medicines and diagnostics, stronger primary health care, more effective public health functions, and improved access to preventive and curative care. These channels can contribute to higher life expectancy over time.

The stronger relationship observed after applying a logarithmic transformation to spending is economically intuitive. At very low levels of public spending, additional resources can finance essential, high-impact interventions and may therefore be associated with substantial improvements in survival. At higher spending levels, the relationship tends to flatten: additional spending may still improve quality, access, and outcomes, but the marginal gains in life expectancy are usually smaller because countries have already achieved high baseline survival.

From a policy perspective, the findings support the importance of adequate, predictable, and efficiently allocated public financing for health. However, the results should not be interpreted as showing that spending alone determines life expectancy. Countries with similar spending levels may achieve different outcomes depending on the efficiency of resource use, the strength of primary health care, financial protection arrangements, purchasing and provider payment systems, governance, health workforce availability, quality of care, and broader social determinants of health.

The 2020 reference year also requires caution. The COVID-19 pandemic affected both life expectancy and public spending patterns, and the direction and magnitude of those effects differed across countries. In some countries, emergency public spending increased sharply, while life expectancy simultaneously declined due to excess mortality. Therefore, the results are best interpreted as a cross-sectional snapshot during an exceptional year rather than as a stable long-term relationship.

5. Key Findings and Implications

- There is a positive cross-country association between government health spending per person and life expectancy at birth in 2020.
- The association is stronger when health spending is assessed on a logarithmic scale, with a log-scale Pearson correlation of 0.823.
- The simple log-linear association suggests that a tenfold increase in government health spending per person is associated with approximately 6.6 additional years of life expectancy, although this is not causal evidence.
- The pattern is consistent with diminishing marginal returns: the largest health gains from additional public spending are likely to occur where baseline spending is low and essential services remain underfunded.
- Policy relevance depends not only on how much governments spend, but also on how funds are pooled, allocated, purchased, monitored, and translated into effective service delivery.

6. Limitations

- The analysis is cross-sectional and does not establish causality.
- The analysis does not control for income, demographic, epidemiological, institutional, or social determinants of life expectancy.
- The year 2020 was affected by the COVID-19 pandemic, which may have distorted both spending and mortality patterns.
- Government health spending per person captures only the public financing component and does not reflect private spending, out-of-pocket payments, service quality, or efficiency.